Eliminative Information in a Generalized Monty Hall Problem

Increasing Marginal Value, Non-Myopic Search, and an Exact Cost Threshold

Abstract

The classical Monty Hall puzzle is usually presented as a one-step probability paradox. This note treats it instead as a small model of costly information acquisition. There are K doors, m prizes, and a knowledgeable host who can sequentially reveal losing doors. After paying for t reveals, the decision maker may switch uniformly to one of the still-closed doors other than the initial choice. The switch success probability is

$$S(t) = \frac{m(K-1)}{K(K-1-t)}.$$

Because host reveals eliminate only losing doors, the marginal value $S(t+1) - S(t)$ is strictly increasing. Consequently the deterministic stopping problem $\max_t\{S(t) - ct\}$ is not governed by a myopic rule. Its optimum is attained at an endpoint: either buy no information or buy all available information. The critical cost is exactly

$$c^*(K) = \frac{1}{K},$$

independent of the number of prizes m whenever at least one reveal is available. Simulations and plots illustrate the resulting phase transition.

1 From Puzzle to Information Acquisition

The standard Monty Hall problem has three doors, one prize, one initial choice, one host reveal, and one option to switch. Its familiar answer, "switching wins with probability 2/3", is not really about luck. It is about constrained information: the host knows where the prize is and is not allowed to reveal it.

We generalize this mechanism. A contestant faces K doors, exactly m of which contain prizes, with $1 \leq m < K$. The contestant first chooses one door uniformly at random. A host then reveals losing doors, never opening the contestant's initial choice and never opening a prize. Reveals are costly: each reveal costs $c > 0$. After t reveals, the contestant switches uniformly to one of the remaining closed doors other than the initial choice.

The maximum number of possible reveals is

$$T = K - m - 1.$$

This is the number of losing doors not chosen by the contestant in the worst case. The feasible stopping times are therefore

$$t \in \{0, 1, \ldots, T\}.$$

Key distinction. The host does not reveal prize values or sample new alternatives. The host only deletes impossible alternatives. This is *eliminative information*. That distinction reverses the usual diminishing-returns intuition.

2 Switching After t Reveals

Theorem 1 (Switching probability)**.** *After t losing doors have been revealed, with $0 \le t \le K-m-1$, the probability of winning by switching uniformly among the available non-initial doors is*

$$S(t) = \frac{m(K-1)}{K(K-1-t)}.$$

The probability of winning by staying is

$$S_{\text{stay}} = \frac{m}{K}.$$

Proof. Condition on whether the initial choice is a prize. With probability m/K, the initial door is a prize. Then among the other $K-1-t$ closed switch options, exactly $m-1$ are prizes. With probability $(K-m)/K$, the initial door is not a prize. Then all m prizes are among the $K-1-t$ switch options. Therefore

$$S(t) = \frac{m}{K}\frac{m-1}{K-1-t} + \frac{K-m}{K}\frac{m}{K-1-t} = \frac{m(K-1)}{K(K-1-t)}.$$

Staying wins precisely when the initial choice was a prize, which has probability m/K. □

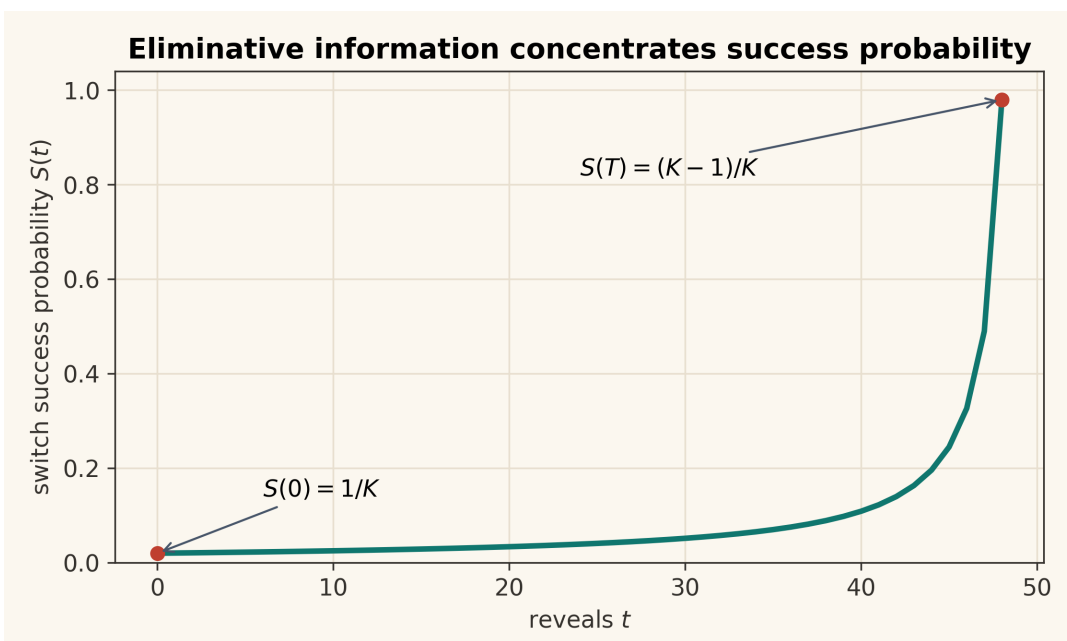

Figure 1: For $K = 50, m = 1$, switch success is almost flat early and then rises sharply as the last losing doors are eliminated.

3 Increasing Marginal Value

Let

$$\Delta_t = S(t+1) - S(t), \qquad 0 \le t \le T-1.$$

Proposition 1 (Marginal value)**.**

$$\Delta_t = \frac{m(K-1)}{K(K-2-t)(K-1-t)}.$$

In particular, Δ_t is strictly increasing in t.

Proof. Subtract consecutive values:

$$\begin{aligned}\Delta_t &= \frac{m(K-1)}{K(K-2-t)} - \frac{m(K-1)}{K(K-1-t)} \\ &= \frac{m(K-1)}{K(K-2-t)(K-1-t)}.\end{aligned}$$

The denominator decreases as t increases, so the marginal value strictly increases. □

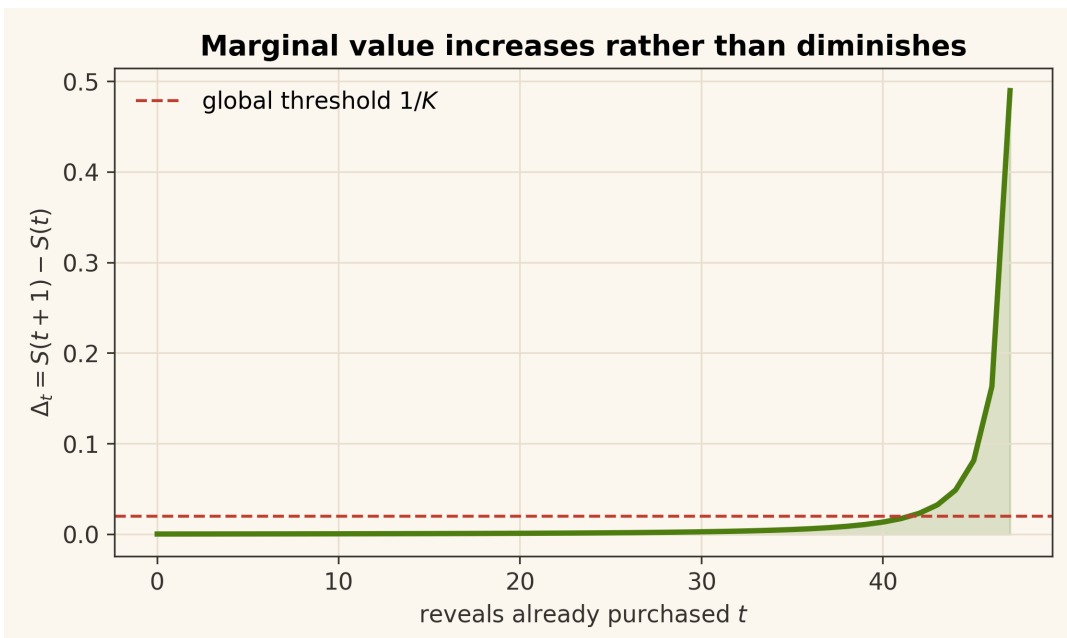

Figure 2: The marginal value of another reveal rises over time. This is the opposite shape from standard diminishing-returns search models.

4 The Costly Stopping Problem

Suppose each reveal costs c, and normalize the value of a prize to 1. After t reveals, switching has net value

$$W(t) = S(t) - ct.$$

The optimal stopping problem is deterministic:

$$t^* \in \arg\max_{0 \le t \le T} W(t).$$

Theorem 2 (Endpoint optimality)**.** *The function $W(t)$ attains its maximum over $\{0, 1, \ldots, T\}$ at an endpoint. Thus every optimal policy either buys no reveals or buys all T possible reveals.*

Proof. The increments of W are

$$W(t+1) - W(t) = \Delta_t - c.$$

Since Δ_t is strictly increasing, these increments cross zero at most once. Hence W first decreases and then increases, or is monotone. Such a discrete convex sequence cannot have a strict interior maximum. Therefore a maximum is attained at $t = 0$ or $t = T$. □

Theorem 3 (Exact critical cost)**.** *If $T = K - m - 1 > 0$, then the endpoint comparison has the exact threshold*

$$c^*(K) = \frac{1}{K}.$$

More precisely,

$$t^* = \begin{cases} T, & c < 1/K, \\ 0, & c > 1/K, \end{cases}$$

and both endpoints are optimal when $c = 1/K$.

Proof. At zero reveals,

$$W(0) = S(0) = \frac{m}{K}.$$

At full revelation, $t = T = K - m - 1$, and

$$S(T) = \frac{m(K-1)}{K(K-1-T)} = \frac{m(K-1)}{Km} = \frac{K-1}{K}.$$

Thus

$$W(T) - W(0) = \frac{K-1}{K} - \frac{m}{K} - c(K-m-1) = (K-m-1)\left(\frac{1}{K} - c\right).$$

The sign is determined exactly by whether c is below, equal to, or above $1/K$. □

5 Why Myopic Search Fails

A myopic policy buys the next reveal only if its immediate gain exceeds cost:

$$\Delta_t \ge c.$$

This rule is safe in many settings with decreasing marginal value. Here it can be badly wrong. For example, with $m = 1$,

$$\Delta_0 = \frac{1}{K(K-2)} \quad \text{but} \quad c^* = \frac{1}{K}.$$

For every cost satisfying

$$\frac{1}{K(K-2)} < c < \frac{1}{K},$$

the first reveal fails the myopic test, yet the globally optimal policy buys all reveals. The early reveals are individually unattractive because their purpose is to unlock the much more valuable late reveals.

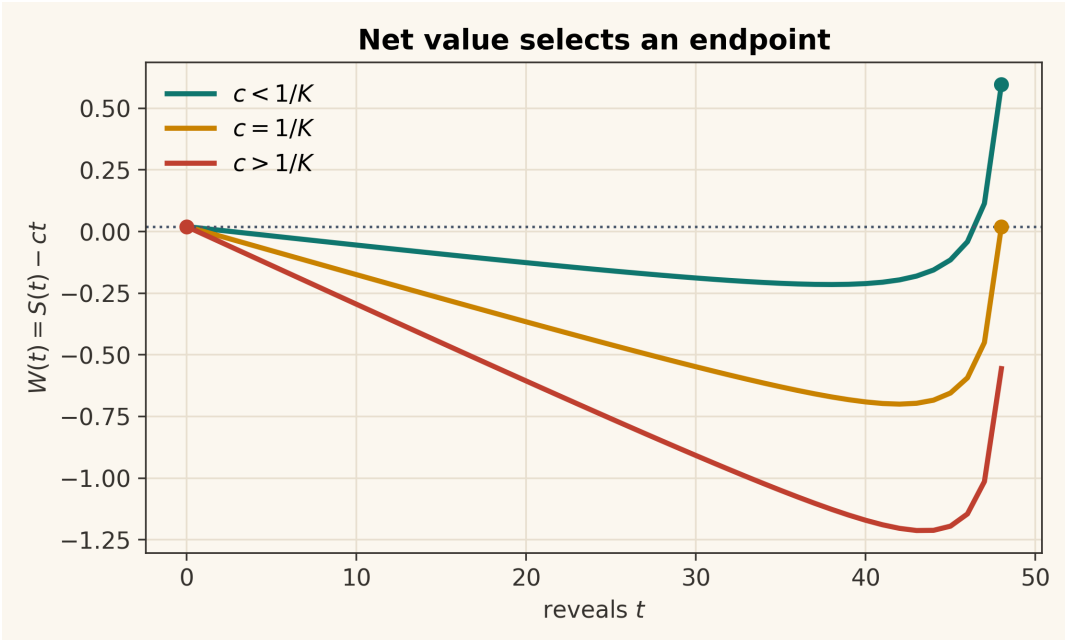

Figure 3: For $K = 50, m = 1$, net value chooses one of the two endpoints. At the critical cost $c = 1/K$, the endpoints tie.

6 Numerical Illustrations

The figures below are not needed for the proof; they make the threshold geometry visible.

7 Relation to Pandora's Box

The problem superficially resembles Pandora's box and other sequential search models: information is acquired over time, and stopping trades off knowledge against cost. The mechanism is different.

Feature	Pandora-style search	Eliminative Monty Hall
Information type	reveals sampled values	removes impossible options
Marginal value	typically decreasing	strictly increasing
Policy intuition	local indices can work	myopic rules can fail
Optimum shape	often interior/reservation	endpoint threshold

The host's knowledge makes opened doors special: they are not random bad draws, but certified non-prizes. Every reveal concentrates the same hidden probability mass over a smaller candidate set.

8 Conclusion

The generalized Monty Hall problem gives a compact example where information has increasing marginal value. The switch probability is explicit, the stopping problem is deterministic, and the optimal policy has an exact phase transition:

$$\boxed{c^*(K) = 1/K.}$$

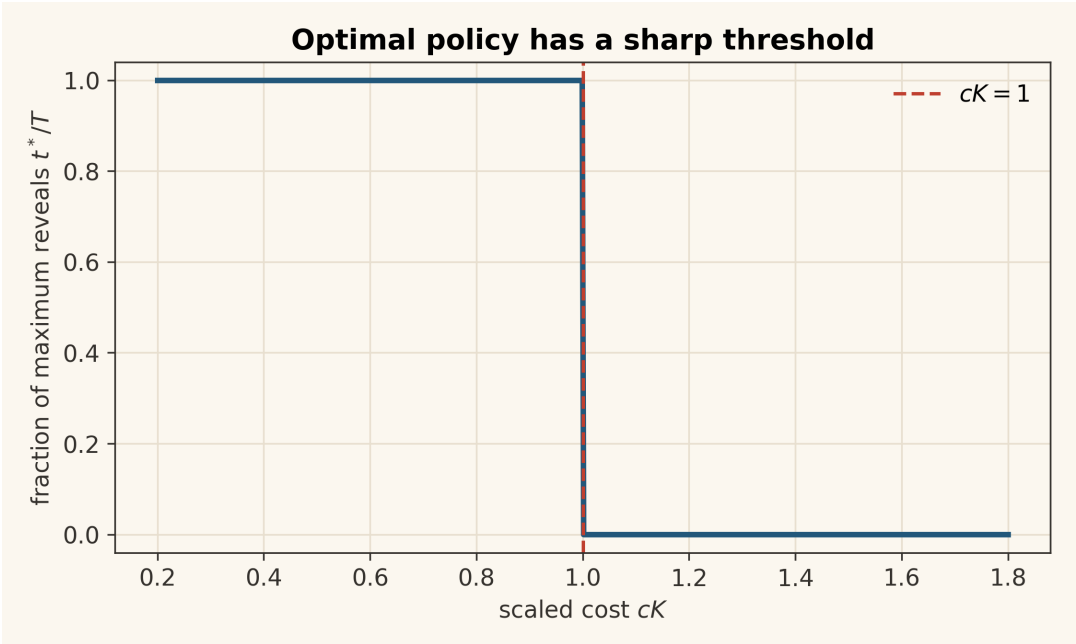

Figure 4: The optimal fraction of available reveals jumps at $cK = 1$.

Below this cost, one should buy every available reveal; above it, one should buy none. The result is a useful toy model for eliminative information: when information removes impossibilities rather than revealing independent values, waiting for late-stage concentration can dominate local reasoning.

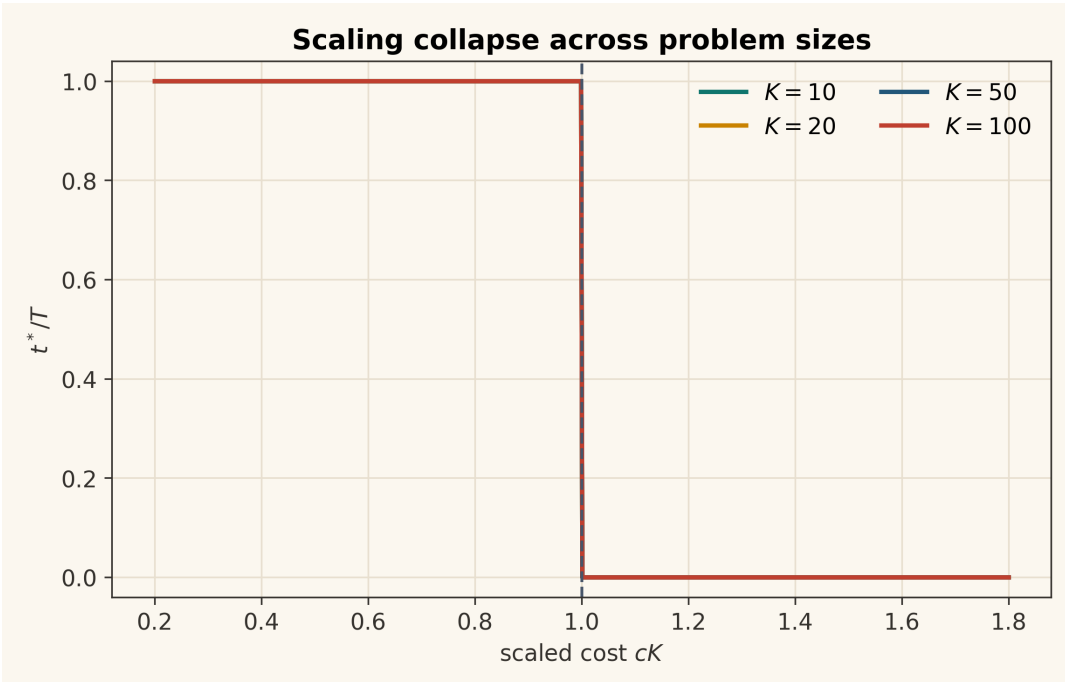

Figure 5: After scaling cost by K, the transition curves collapse across problem sizes.

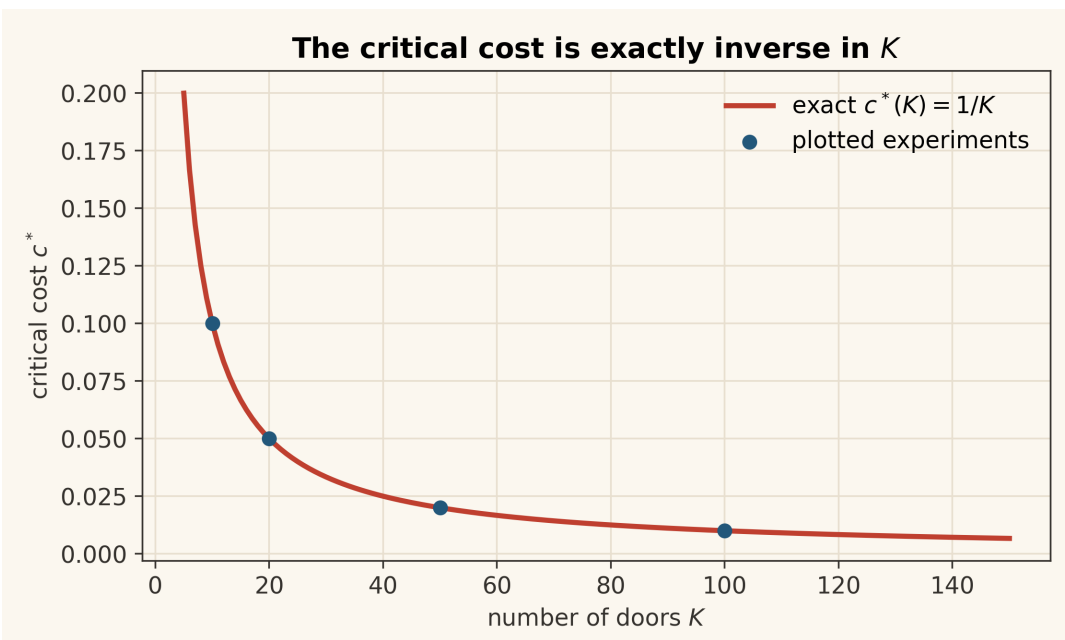

Figure 6: The critical cost is exactly $1/K$; the numerical experiments sit on the theoretical curve.